

point-detection lineage



Radar & point datasets

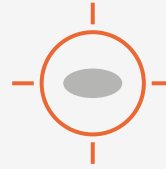
RADIATE tiny_foggy
real automotive radar in fog
synthetic point generator
controlled tracks and clutter

image lineage



Image datasets

PhC-C2DL-PSC
real cell microscopy, 300 frames
SyntheticMHTData.py
synthetic frames, tunable noise



Find the blobs

turns pixels into points:
background subtraction, then
connected components

one puzzle per cluster of tangled tracks

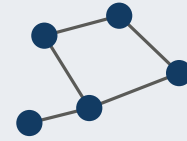
Shared classical MHT front end



detections

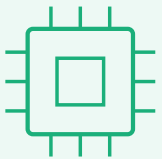


candidate tracks



conflict graph

same puzzle, two solvers



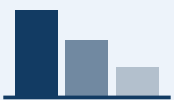
Ordinary computer

solves the graph exactly:
MILP / max-weight clique,
then lists the top- K best
compatible sets of tracks



Atoms & lasers

one atom per candidate track;
clashing tracks sit close, so the
Rydberg blockade forbids both.
Laser sweep, then many shots



Ranked global hypotheses

the best set of compatible tracks, plus the runners-up,
each with its probability $P\{H_j\}$ and $P\{T_i\}$

the two solvers are compared on: which answer they find,
the cost per answer, and whether the graph could be embedded